200 Water Rooftop Virtual Showroom

Technical overview of the interactive 3D rooftop experience, based strictly on project files.

Purpose & Experience

- Creates a scroll-driven 3D building tour that transitions into a roof-only inspection mode for rooftop systems.
- Users open the detail view, rotate the roof, and inspect callouts, leader lines, legend, and flow overlays.

Technology Stack

Runtime	Static HTML/CSS/JavaScript; Tailwind CDN; Lucide; Roboto; no build step documented.
3D + motion	Three.js 0.170, GLTFLoader, MeshoptDecoder, WebGLRenderer, PerspectiveCamera, GSAP ScrollTrigger.
Models	Full building: logos/200_water.opt.glb. Roof detail: logos/200 water Roof Only.glb.

Engineering Workflow

- model-preload.js centralizes GLB loading: building first, roof deferred and prefetched later.
- The building GLB is cloned, material-prepared, centered, uniformly scaled, rotated, and framed with a custom camera/light rig.
- Rooftop framing samples upper-roof vertices and adjusts projection until the inspection view fits cleanly.
- Roof-only alignment replays the building transform, then screen-aligns the separate roof model.

Code evidence: deferred model loading

```
export const buildingGltfPromise = preloadGltf(paths.building);
if (!roofGltfPromise) roofGltfPromise = preloadGltf(paths.roof);
```

Interaction Engineering

Scroll choreography	A smoothed playhead synchronizes camera, FOV, lights, CTA state, render timing, and roof prefetch.
DOM + WebGL sync	rooftop-labels.js projects 3D anchors so DOM labels, leader lines, legend, and SVG flow paths stay aligned with the model.

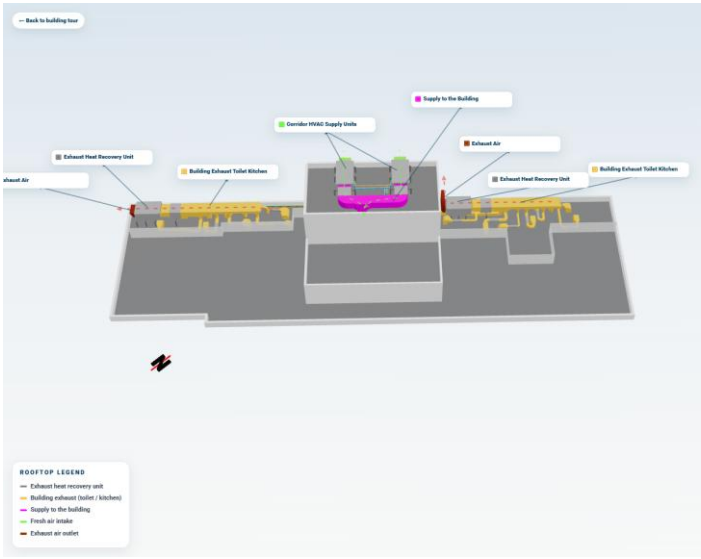
Performance Engineering

- Primary GLB is preloaded and cached; roof loading is deferred until needed.
- MeshoptDecoder handles compressed GLBs; optimized metadata includes EXT_meshopt_compression/KHR_mesh_quantization.
- Renderer DPR is capped, mobile antialiasing is disabled, and rendering pauses outside the active 3D section.

Architecture

User scroll / CTA -> GSAP ScrollTrigger + smoothing -> Three.js scene/camera/lights -> full-building GLB and roof-only alignment -> DOM/SVG labels and flow overlays

Rooftop Detail View



Rooftop interaction shown

- Roof-only inspection state loaded from the project rooftop GLB.
- Equipment labels, leader lines, legend, and flow overlays presented in one coordinated view.
- Screen-space DOM/SVG annotations aligned to the underlying WebGL model.

Figure: active rooftop detail view captured from the project implementation.

Most Complex Implementation Areas

Cross-model alignment	Transform replay plus projected-bounds matching bridges full-building and roof-only GLBs with different coordinate assumptions.
Rooftop camera math	Roof vertex sampling, projection checks, and iterative distance fitting create the inspection composition.
Overlay precision	DOM labels and SVG flows remain attached to live 3D anchors through resize, drag, and model swapping.

Code evidence: scroll smoothing

```
const delta = targetProgress - playhead.progress;
playhead.progress += delta * smoothing;
timeline.progress(playhead.progress);
```

Evidence & Limits

Evidence reviewed: runtime HTML/CSS/JS modules, README, building-scroll-recipe.md, wrangler.jsonc, audit scripts/screenshots, and GLB metadata from logos/*.glb.

- Blender is verified only as source/export evidence; active optimized GLBs report glTF-Transform v4.4.2.
- Not verified: source .blend file, backend code, package/build pipeline, or live import/use of rooftop-retrofit.js.